

FinPersona-Bench

Consolidated Three-Reviewer Audit Report

Independent review of the current paper, the 47-paper reference corpus, and the Revision Plan for Resubmission

Field	Value
Date	20 August 2026
Paper audited	docs/COLM_FinPersona_Bench_Current_Paper.pdf (29pp, App. A-L)
Version note	Verified byte-identical in text to COLM_FinPersona_Bench_Post_Rebuttal.pdf (similarity 1.0000, zero change blocks)
Plan audited	docs/FinPersona-Bench_ Revision Plan for Resubmission.docx
Reference corpus	47 papers: 36 in references/related_work_2026 and references/mech_interp; 11 in references/gap_sweep_aug2026
Reviewers	Three independent agents, distinct assigned priors, no shared context
Verification	All load-bearing claims re-checked by the orchestrator against paper text and repository source
Purpose	Input to a further review cycle prior to implementation

Reviewer stances were assigned deliberately so that conclusions could be triangulated rather than echoed. None of the three saw the others' work, and none was given the orchestrator's prior analysis.

Reviewer	Assigned stance	Primary deliverable
A	Priority auditor - assumes no claim is novel until verified against the corpus	Claim-by-claim prior-art audit of 19 contribution claims
B	Methods and construct-validity adversary	Metric, causal-design and statistical audit
C	Program-committee strategist	Competitive position map and rebuilt revision plan

Executive Summary

Three independent reviewers converged on eleven substantive findings. The most consequential was discovered by Reviewer C in the repository source code and independently inferred by Reviewer B from the paper's own equations, and has been confirmed by the orchestrator:

The benchmark's agents are memoryless. Context does not accumulate in either experimental arm. The paper's titular mechanism - Mandate Salience Decay under accumulating context - is therefore untested by its own design, and the M1-M4 mechanistic programme in the revision plan cannot be run as specified.

Two further findings of similar severity were confirmed:

- The hidden fundamental value is exactly recoverable from an observable feature the agent reads at every step. Reported $P/E = P_t/EPSt$ with $EPSt = V_t/15$ gives $P/E = 15 \cdot P_t/V_t$ exactly; dividend yield gives a second exact disclosure. The benchmark's stated core technical contribution does not hold as implemented.
- The 4.4x compounding headline is an artifact of a cumulative statistic. Reviewer C located `expanding().min()` in the analysis code and showed the gap peaks at Q3 and falls at Q4, vanishing to 0.0003 when cumulation is removed. Reviewer B independently reproduced a 2.59x compounding gap from two perfectly constant exposures.

Notwithstanding, all three reviewers independently identified the same surviving contribution, and it is genuinely unclaimed in the corpus:

Holding environment, injection position, cadence and model fixed, and varying only the semantic content of the injected mandate, the sign of the intervention's behavioural effect reverses - 17/18 models improve under a capital-preservation mandate and 16/18 degrade under a growth mandate, in an identical market.

The consensus verdict is that the paper is not acceptable in its current form, that the revision plan allocates its budget to the wrong programme, and that a substantial fraction of the required work costs no compute at all.

Independent Verification Register

Reviewer findings are agent output and were not taken at face value. The following claims were re-checked directly against the paper text or the repository source by the orchestrator. Every claim listed was confirmed.

#	Claim	Raised by	Evidence located	Status
1	No accumulating context in either arm	B (inferred), C (code)	grep of agent/ returns zero MessagesPlaceholder, chat_history, conversation_history or message-append constructs	CONFIRMED
2	temperature = 0.2, not 0.0 as stated in App. D	C	static_agent.py:57, ocean_static_agent.py:55, legacy_agent.py:51	CONFIRMED
3	Hidden fundamental is exactly recoverable	B	Table 2 line 1126: $P/E = Pt/EPSt$, $EPSt = Vt/15$; line 1129 dividend yield	CONFIRMED
4	RG saturated by 'buy one share then HOLD'	B, A	Line 443: $Ht = \max(0, Wt - Ct * Wt) = Wt(1 - Ct)$; the $\max(0, .)$ clamp is vacuous	CONFIRMED
5	RG's stated 0.5 random baseline is wrong	B, A	Line 452; true baseline is approx. 0.667 given the HOLD clause	CONFIRMED
6	Figure 5 crash series sign-inverted vs Table 11	A	Fig 5: -66.0, +11.3, +59.1. Table 11: +66.0, -11.3, -59.1. Flat and bull-trap columns match	CONFIRMED
7	Crash result fails at model level	B	Table 11 crash column 12 positive / 6 negative; two-sided sign test $p = 0.238$ vs reported 0.002	CONFIRMED
8	Table 1 and Table 9 sign conventions contradict	B, A	Lines 538 and 1936	CONFIRMED
9	Placebo range misstated	A	Paper states 0.003-0.045; ISFJ is $0.950 - 0.903 = 0.047$	CONFIRMED
10	Bull-trap 'stabilize' claim contradicts App. A	A	Line 684 vs App. A tau-3 'Blow-off Top: price disconnects exponentially'	CONFIRMED
11	Paper cites none of the 47 corpus papers	A	Zero hits for Arike, Dongre, Laban, Kocielnik, Tosato, Byrd, ABIDES, Hashimoto, Menon, Ross	CONFIRMED

Part I - The Architecture Finding

I.1 The agents have no memory

agent/static_agent.py lines 101-103:

```
self.prompt_template = ChatPromptTemplate.from_messages([
    ("system", "{persona}"),
    ("human", "{input_data}\n\nIMPORTANT: You must return a valid JSON object...")
])
```

agent/memory_agent.py lines 65-67 is identical plus a {context_refresh} slot. reset() is a no-op because there is no state to reset. A grep across agent/ for MessagesPlaceholder, chat_history, conversation_history, self.history and message-appending returns nothing.

Each of the 200 trading days is an independent single-turn API call with a constant-length prompt. The only quantity crossing steps is the portfolio tuple computed by PortfolioTracker.

I.2 Consequences

1. The paper's central mechanism cannot occur. The abstract, Figure 1, section 1, section 3.2 and the Conclusion all assert that mandates erode as market context accumulates. No context accumulates. The mandate sits at the same token distance from the generation position on day 200 as on day 1.
2. M1-M4 are unrunnable as specified. M1 proposes logging attention mass on the mandate span across the rollout. There is no rollout; the trajectory would be flat by construction. M2, M3 and M4 inherit the problem.
3. The memory arm is a misnomer. Both arms carry the mandate in the system prompt at every step. The manipulation is 'mandate appears once versus twice within a single prompt, at two positions'. That is prompt composition, not memory.
4. Sampling variance is uncontrolled. Appendix D claims deterministic decoding at temperature 0.0; the code sets 0.2 and passes no top_p. The five seeds reseed only the market generator, yet all 270 pairs enter one Wilcoxon as independent observations.

Reviewer C's assessment: this single fact invalidates the paper's mechanism claim and the plan's entire mechanistic programme simultaneously. Reviewer B reached the same conclusion from Eq. 3 and Eq. 5 alone, noting that the paper describes the static agent as a 'stateless predictor' (line 324) while simultaneously claiming it is 'susceptible to context drift as immediate market observations overwhelm initial instructions' (lines 330-332) - an internal contradiction.

Part II - Reviewer A: Priority Audit

Method: 19 distinct contribution claims extracted from the abstract, contribution list, metric definitions and every appendix experiment. Each graded against the corpus as NOVEL, PARTIALLY ANTICIPATED, FULLY ANTICIPATED, or ANTICIPATED AND CONTRADICTED.

II.1 Verdict summary

Claim	Verdict	Anticipating work
C1 Formalising MSD	FULLY ANTICIPATED	Meta 2607.08716 defines 'behavioral state decay' near-verbatim (Jul 2026); Kim/Li COLM 2024 'instruction drift'; Arike goal drift
C2 Behaviour distinct from reasoning	FULLY ANTICIPATED	DriftBench KBV 8-99%; CLQT coherence gap +0.30/+0.23; Everitt goal-directedness vs performance; Kocielnik
C3 Decoupled synthetic ground truth	PARTIALLY ANTICIPATED	Hashimoto (hidden GBM fundamental, LOB, 500 days); Machine Spirits ($p_f = 60$); ABIDES oracle
C4 Three failure-mode regimes	PARTIALLY ANTICIPATED	Regime-switching design is standard; novel only as a psychometric partition
C5 MAS	PARTIALLY ANTICIPATED + COMPROMISED	CLQT a_target 'mandate alignment'; Arike GD_actions budget-share deviation
C6 Caricature Index	NAME ANTICIPATED, CONSTRUCT INVALID	Deflanderization names flanderization; CI is Maximum Drawdown
C7 Rationality Gap	PARTIALLY ANTICIPATED + BROKEN	ABM fundamentalist component; Machine Spirits RD/RDMAX
C8 Model-dependence	FULLY ANTICIPATED	ContextEcho 23 models; PERSIST 25 models; DriftBench 7 models; Arike Finding 4
C9 Compounding / 4.4x	ANTICIPATED AND CONTRADICTED	Drift No More 2510.07777 bounded equilibria, restoring force $\rho < -0.7$. Uncited.
C10 Re-grounding not universally beneficial	PARTIALLY ANTICIPATED	Meta always-on vs selective ablation; ContextEcho mode-dependent sign flip; Kim/Li cost curve; CLQT no-memory ablation
C11 Placebo control	FULLY ANTICIPATED	ContextEcho filler arm in every cell (41,921 evals); When Attention Closes App. H identical 3-arm design; Arike token-distance ablation
C12 Big Five and O3 ablation	PARTIALLY ANTICIPATED	Kocielnik; Song; PERSIST; Ross and Lo risk tolerance 57-88% of predictive weight
C13 Injection-frequency ablation	FULLY ANTICIPATED	Kim/Li COLM 2024 swept injection probability p with an MMLU cost axis
C14 T-calibration	NOVEL but not a contribution	No prior work; internal power analysis, $N = 9$, most cells never significant
C15 Crash discount sensitivity	NOT A CONTRIBUTION; inverts the mechanism	Memory benefit is largest in the mildest crash

C16 LCR / rationale analysis	NOVEL construct, falsified internally	DriftBench surface fidelity gap; CLQT coherence gap
C17 MBTI-as-vocabulary and title	ANTICIPATED and undercut	Kocielnik; Song; PERSIST
C18 Selective re-grounding prescription	FULLY ANTICIPATED	Meta built it; CLQT section 3.5 ships a Ct < 0.7 trigger
C19 Transfer to triage and legal	ANTICIPATED - already executed	Menon runs ER triage

II.2 What survives, in the exact narrow form

S1 - Content-isolated sign reversal (strongest, defensible).

In a fixed decision environment, with injection position, cadence and model held constant, varying only the semantic content of the re-injected mandate reverses the sign of the intervention's behavioural effect. Does NOT survive in the broad form 're-grounding is not universally beneficial', which is settled.

S2 - Longest-horizon ground-truthed instrument (strong, artifact claim).

200-step and 800-step paired grid, 18 models by 3 mandates by 3 regimes by 5 seeds, scored against a generating function rather than consensus or an LLM judge. Arike approx. 30 steps; When Attention Closes 50 turns; Drift No More approx. 10 turns; DriftBench 6 turns; ContextEcho reaches 9,716 turns but has no behavioural task and no ground truth.

S3 - The two-sided content test (moderate, under-powered).

Pairing a subtractive control (placebo) with a reductive control (O3 vocabulary stripping) on the same environment and metric. The pairing is unclaimed; the execution is 15 pairs, one model, one scenario, with both key contrasts non-significant.

S4 - Capacity-conditioned crash reversal (weak).

Mini variants reverse where flagships do not. Broken by GPT-5-Mini (+5.1), by the LCR counterexample, and by DriftBench's opposite capacity ordering.

Part III - Reviewer B: Methods and Construct Validity

III.1 The unifying defect - three metrics, one variable

Portfolio value is $W_t = C_t * W_0 + (1 - C_t) * W_0 * (P_t / P_0)$. Writing equity exposure $e = 1 - C$:

- MAS is a direct function of C_t .
- CI is monotone in e and identically zero for an all-cash agent.
- RG's only clause that discriminates static from memory is HOLD and $P \leq V$ and $C_t = 1$, which is a cash-level test.

A single intervention that raises mean cash therefore lowers MAS for ISFJ, raises it for ENTJ, shrinks the magnitude of CI in the crash, and lowers RG in the bull trap - precisely the reported result set in precisely the reported signs.

Table 9 reports the underlying variable as a secondary metric: average cash 35.6% to 49.4%, +38.8%, $p < 0.001$ - the largest and most significant effect in the paper. The paper never regresses out cash level and never tests whether the three metrics carry independent information. CLQT reports an inter-axis correlation matrix (mean absolute $r = 0.23$) precisely because orthogonality must be demonstrated rather than assumed.

III.2 MAS

- The target contradicts the mandate. ISFJ is instructed to prefer index funds, bonds and sector leaders, and to buy puts and hedges; $C_{ideal} = 1.0$ scores 100% cash as perfect adherence. The prompt says buy things; the metric says buy nothing.
- The cited theory contradicts the targets. With μ approx. 0 in the flat scenario, Markowitz gives w^* approx. 0, so C_{ideal} approx. 1.0 for every persona in the only scenario where MAS is computed.
- The empirical literature contradicts the magnitude. Jiang, Peng and Yan (JFE 2024) yields adjusted R-squared of 0.05 and a 5-8 percentage point equity swing across the full Big Five range. The paper posits an 80-point spread, using an instrument that structurally cannot encode Neuroticism - the trait with the largest coefficient.
- A broken agent scores perfectly. $MAS = 0.000$ over 200 days means the agent never traded. That is indistinguishable from parse failure, refusal, or an always-HOLD default. No parse-failure or refusal rates are reported anywhere in the paper.
- Cross-persona incomparability. Ranges are persona-specific (ENTJ 0 to 0.8, ISFJ 0 to 1.0). Pooling lets the ISFJ scale swamp a sign reversal present in 16 of 18 models; Table 11's flat column shows only 3 of 18 negative as a result.

Worked counterexample: an ENTJ that buys to 80% equity on day 1 and never re-evaluates scores $MAS = 0.000$. An ENTJ that reads the mandate daily, observes no trend, and correctly declines to chase scores 0.700. The metric ranks the automaton above the reasoner.

III.3 Caricature Index

The construct is persona-relative, signed, and a change measure. Maximum Drawdown is persona-agnostic, unsigned, and a level measure. An ISFJ that caricatures itself under stress liquidates to cash and posts a drawdown of exactly 0 - the purest instance of the construct scored as its complete absence.

CI and RG are normatively opposed by construction. Appendix A enforces $P \leq 0.98V$ for all $t \geq \tau-1$ in the crash, so BUY is value-rational throughout; maximum exposure therefore maximises RG and minimises CI in the same scenario. The paper never reconciles this.

III.4 Rationality Gap

- $H_t = \max(0, W_t - C_t * W_t) = W_t(1 - C_t)$, so $H_t > 0$ whenever the agent holds any stock at all. The $\max(0, \cdot)$ clamp is vacuous.
- Degenerate policy: buy one share on day 1 and HOLD for days 2-200 gives RG approx. 1.000. A one-line rule saturates the headline metric of the bull-trap failure mode.
- The stated range is wrong. A uniform-random agent holding stock scores 2/3, not 0.5, so reported values of 78.4 and 85.9 must be read against 66.7 rather than 50.
- The ground truth is not hidden. $P/E = 15 * P_t/V_t$ exactly; dividend yield gives a second exact disclosure. The constant 15 is an unanalysed free parameter that determines the metric.
- No baselines of any kind are reported - no buy-and-hold, constant-mix, random, or always-HOLD reference against which the 78-86 range could be interpreted.

III.5 Causal and experimental design

What the placebo does not rule out:

5. Any imperative instruction at the recency position. The placebo is declarative boilerplate; the treatment is a capitalised second-person command with an attention-grabbing delimiter. The arms differ in imperative force, not only in mandate content.
6. The bidirectionality argument is invalid. A uniform effect on cash level produces a bidirectional effect on MAS because C_{ideal} is monotonically ordered by persona. Any content-free intervention that makes models more cautious generates the entire persona-scenario alignment signature.
7. Position and repetition are confounded. Kim/Li published the isolating condition as System Prompt Repetition in 2024 and treated it as a baseline, with a mandatory capability-cost axis this paper lacks.
8. The placebo's own result cuts against the paper. Static versus Placebo is significant ($p = 0.041$) and the placebo is worse than static for all three personas - direct evidence that injecting any text at the recency position degrades adherence uniformly.
9. Static versus Memory is not significant in the appendix's own data ($p = 0.890$).

The decisive missing control is a swapped mandate: inject the ISFJ mandate into an ENTJ agent and vice versa. If effects track injected content, MAS should move toward the injected target. Cost: 15-90 simulations.

III.6 Statistical audit

Effective N is 18, not 270. The same five seeds are reused across every model and persona.

Recomputing at model level:

Metric	Sign split	Model-level p	Paper's p
Flat MAS	15 / 3	0.0075	0.028
Crash CI	12 / 6	0.238	0.002
Bull-trap RG	3 / 15	0.0056	< 0.001

The crash result - the flagship, the source of the 4.4x claim and the panic-selling failure mode - does not survive treating the model as the unit of analysis.

Under Holm correction within Table 9's own seven-test family, MAS fails family-wise error control (adjusted $p = 0.084$) though it survives Benjamini-Hochberg (0.039). It is the weakest of the five significant results despite being the headline.

A hard constraint governing several appendices: the minimum attainable two-sided Wilcoxon p is $2 / 2^n$. At $n = 3$ (Appendix H) that is 0.250 - no result can ever be significant. At $n = 5$ (per-persona cells in Appendix F, all of Appendix G) it is 0.0625 - no per-persona or per-model claim in those appendices can reach $\alpha = 0.05$ even in principle.

Appendix E reports Claude Sonnet 4.6 in the flat scenario at $p = 0.820$ - and that exact cell is the basis for the placebo control, the Big Five validation, the frequency ablation and the $T = 800$ run.

III.7 Other defects

- No effect sizes or confidence intervals anywhere. The MAS effect is 0.049 against a standard deviation of approximately 0.20.
- Appendix H reports $n = 600$ rationale steps when the real sampling unit is 3 seeds - an inflation of 200x.
- Appendix H recommends $k = 1$ for ENTJ on a linguistic metric while its own text states that $k = 1$ worsens ENTJ's MAS from 0.202 to 0.329.
- The DistilBERT persona classifier is trained on days 1-5 rationales from the same runs it evaluates, and may detect mandate-vocabulary echo rather than the construct. It is load-bearing across Appendices F, H and L and the open-source rebuttal results.
- Table 10 and Tables 1/9 disagree at the default discount (gap 16.5% versus 12.6%), explained only by a 14- versus 18-model difference that is never stated as the reason.

Part IV - Reviewer C: Strategic Position and Plan Feasibility

IV.1 The 4.4x is a cumulative-statistic artifact, and non-monotonic

analysis_may/numerical_analysis.py:

```
df["CI_rolling"] = df["Drawdown_t"].expanding().min()      # cumulative
df["MAS_rolling"] = df["MAS_t"].expanding().mean()         # cumulative
df["RG_rolling"] = df["yt"].expanding().mean() * 100      # cumulative
```

expanding().min() is monotonically non-increasing, so any constant per-step difference accumulates into a growing gap. Section 4.3.2's stated falsification test - that a fixed difference in risk tolerance would produce a constant offset - is mathematically false for this statistic.

Recomputed from results_april crash runs at discount 0.92 (510 runs):

Quartile	Paper's metric (expanding-min)	Quartile-local (peak reset)
Q1	0.0100	0.0100
Q2	0.0245	0.0261
Q3	0.0577 (peak)	0.0446
Q4	0.0424	0.0003

Two results. Under the paper's own metric the gap peaks at Q3 and falls in Q4 - it does not grow monotonically to 4.4x. And removing cumulation makes the Q4 gap vanish. The divergence tracks the scripted panic phase (days 81-140) and disappears during scripted stabilisation. Reviewer B independently reproduced a 2.59x gap from two perfectly constant exposures.

IV.2 The flat-market drift is a portfolio ratchet

Persona and arm	Cash fraction Q1 to Q4	P(BUY) Q1 to Q4
ISFJ static	0.657 / 0.440 / 0.354 / 0.301	0.12 / 0.05 / 0.06 / 0.05
ISFJ memory	0.847 / 0.766 / 0.732 / 0.691	0.05 / 0.03 / 0.04 / 0.04
ENTJ static	0.439 / 0.323 / 0.275 / 0.267	0.22 / 0.21 / 0.22 / 0.22
INTJ static	0.487 / 0.253 / 0.165 / 0.117	0.14 / 0.07 / 0.10 / 0.08

For ISFJ, MAS rises 107% - called mandate decay - while the per-step probability of the mandate-violating action falls 58%. For ENTJ the BUY rate is flat at 0.22 across all four quartiles while cash falls 39%. Cash depletes because each buy converts cash to equity and almost nothing converts back.

The instantaneous propensity to violate the mandate does not decay. MAS integrates a stationary-to-declining policy and trends anyway. The ISFJ static-memory cash gap across days 1 to 200 is 0.012, 0.079, 0.098, 0.215, 0.285, 0.364, 0.393, 0.397 - concave and saturating. Decay predicts acceleration; the data decelerate.

IV.3 Competitive position map

Axis	Owner	FinPersona position
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Goal drift over long horizons	Arike 2505.02709; Menon 2603.03258; Everitt 2504.11844	Behind - three papers refute the mechanism
Longitudinal persona evaluation	ContextEcho 2605.24279 (23 models, 9,643 turns)	Badly behind - 200 memoryless calls is not 200 turns
Synthetic market, hidden fundamental	Hashimoto 2510.12189; Machine Spirits 2604.18602; ABIDES	Level; narrowly ahead only on V as scoring criterion
Re-grounding interventions	Meta 2607.08716; CLQT 2606.29771 s3.5; ContextEcho	Level to ahead on content by regime sign reversal only
Persona and psychometric framing	Kocielnik 2606.12730; Song 2509.10078; PERSIST	Behind - framing actively contradicted
Financial agent benchmarking	CLQT 2606.29771; Ross and Lo 2604.23837	Level; ahead on multi-persona by multi-regime grid

Note: CLQT section 3.5 already ships a threshold-triggered mandate re-grounding controller in a financial benchmark ('When $C_t < 0.7$ a drift warning is injected'). But CLQT never defines a `_target`, never ablates the investment-target module, and names varying the mandate as future work in section 8.8. That is the remaining seam.

IV.4 Recommended reframing

Proposed title: 'When Reminding Backfires: Mandate Re-Grounding Is Bidirectional in LLM Financial Agents'.

Remove from the abstract: 'Mandate Salience Decay', 'as market context accumulates', 'compounds over time', '4.4x', 'psychometric'.

Being single-turn should be reframed as an asset rather than concealed as a defect: it makes the effect cleanly attributable to one prompt element, cheap to replicate, and directly relevant to the dominant stateless-agent deployment pattern.

Part V - Convergence Across All Three Reviewers

Three reviewers with different assigned priors, no shared context and no access to the orchestrator's prior analysis reached identical conclusions on eleven points. This is the strongest signal in the exercise.

#	Convergent finding	A	B	C
1	The mechanism is untested - no context accumulates	yes	yes (from Eq. 3/5)	yes (from source code)
2	The 4.4x compounding claim is an artifact	yes (quartile/phase confound)	yes (simulation)	yes (expanding().min())
3	Caricature Index is Maximum Drawdown and measures no caricature	yes	yes	yes
4	RG is broken - vacuous clause, trivially saturable, wrong range	yes	yes	yes
5	C_ideal is indefensible; JFE derivation will shrink or destroy the effect	yes	yes	yes
6	Effective N is 18, not 270; the crash result does not survive	yes	yes (quantified)	yes
7	The placebo confounds semantic content with directive force	yes	yes	yes (verified in code)
8	Cut M1, M3, M4	M3 conditional	yes	yes
9	Mandate wording sensitivity must be promoted from LOW to top-three	yes	yes	yes
10	Retitle - 'psychometric stability' is unsupportable	yes	yes	yes
11	The surviving contribution is the content-isolated sign reversal	yes (S1)	yes	yes

Part VI - Divergence

On M3 (causal ablate and restore).

Reviewer A calls it the one mechanistic item worth doing - the only item that answers Reviewer VG7b's confound objection without moving prompt tokens - but demands a pre-registered null protocol. Reviewers B and C both say cut it, citing the well-powered null in When Attention Closes Appendix D.5 (all permutation $p > 0.05$; true-patch and random-patch flipping the same pairs in lock-step). Recommended resolution: A is right that it is the only clean answer to the objection, and B and C are right about the odds. Run it last, pre-registered, budgeted for a null.

On C_ideal.

B recommends abandoning fixed targets entirely in favour of Ross and Lo's input-sensitivity criterion (feature concentration over an orthogonally sampled profile space). C recommends keeping targets

but adding a $t = 0$ persona-separability baseline. A recommends deriving from the JFE literature but reporting stipulated and derived side by side. All three agree the current targets are indefensible and that naive derivation backfires.

On whether the sign reversal is genuinely content-driven.

B argues it may be a generic cash shift refracted through oppositely-signed targets. A and C treat it as the surviving contribution but flag the tautology risk: the ENTJ mandate literally instructs 'If the trend is up, BUY', MAS penalises trading, and the O3 ablation removes that instruction and the penalty vanishes. The three converge on the resolution: the phenomenon is real, the content-attribution is unproven, and the swapped-mandate arm settles it.

Part VII - Consolidated Verdict

VII.1 What is the unique contribution now

Holding environment, injection position, cadence and model fixed, and varying only the semantic content of the injected mandate, the sign of the intervention's behavioural effect reverses - 17/18 models improve under a capital-preservation mandate and 16/18 degrade under a growth mandate, in an identical market.

Nobody owns this. Meta varies policy (when to inject). ContextEcho varies deployment mode. Kim/Li vary intervention strength. CLQT varies module presence. FinPersona varies only the mandate text against a fixed environment.

Secondary and weaker: the 200-800 step paired grid scored against a formal generating function is the longest ground-truthed mandate-adherence instrument in the corpus - though 'ground truth' must be restated, since P/E discloses V exactly.

VII.2 Where we stand

Not acceptable as written. All three reviewers reached this independently.

The decisive risk is not scientific but forensic: the paper ships with a Code link, and a reviewer who opens the repository finds the memoryless agents and `expanding().min()` within twenty minutes. Combined with zero citations to any of the 47 nearest neighbours - including a COLM 2024 paper that already ran the frequency ablation this submission proposes as future work - this reads as a soundness failure plus inadequate literature control.

Against the three Program Chair rejection reasons:

PC reason	Plan status	Assessment
Focus on MBTI	Partially on track	Rename is marked LOW and should be HIGH; OCEAN scaling is framed as validation, which Kocielnik contradicts; C_ideal derivation will backfire
Limited generalization to real markets	Partly on track	Diagnosis correct - the 'realism reintroduces subjectivity' defence is false and must be deleted. The ABIDES citation is mis-specified; Hashimoto is the correct existence proof
Design of the memory test	NOT on track	The plan treats this as a causal-identification problem for M3. The actual defects are prompt-level and fixable at a fraction of the cost; M3 fixes none of them

VII.3 What must change in the revision plan

The plan's diagnosis is right and its prescription is wrong. It spends its entire budget on an interpretability programme that cannot run on this architecture, is twice-published where it can run, and has documented nulls where it would be novel.

Part VIII - Prioritised Programme

Tier 0 - Zero compute, one week, do first

If Tier 0 fails, everything downstream changes. All items are re-analysis of existing data.

10. Cash-mediation analysis: regress MAS, CI and RG on mean cash fraction; report condition effects controlling for cash.
11. Metric correlation matrix across all 270 cells (Reviewer B predicts absolute $r > 0.7$ for at least three pairs).
12. Recompute the compounding claim without expanding(); fit $\Delta D_t = a + b \cdot D_t$ and report the implied equilibrium.
13. Report the memory-minus-static gap at $t = 1$ and its growth curve.
14. Degenerate-behaviour audit: parse failures, refusals, zero-trade runs, and a 99% interval on C_t per cell.
15. Mixed-effects re-analysis with random effects for model, persona and seed; Benjamini-Hochberg correction; effect sizes and confidence intervals throughout.
16. Trivial-policy baselines: buy-and-hold, always-HOLD, random, constant-mix at C_{ideal} , and the rule 'BUY if $P/E < 15$ else SELL'.
17. Fix Figure 5's crash signs; harmonise the three sign conventions; correct the placebo range; fix the section 4.4 self-reference; define the dagger in Table 11; reconcile the Sonnet 4.6 values across Appendices E, F and H.
18. Add the citations and write the differentiation paragraphs. Minimum set: Meta 2607.08716, When Attention Closes 2605.12922 (including Appendix H), Kim/Li COLM 2024, Drift No More 2510.07777, Arike 2505.02709, Menon 2603.03258, CLQT 2606.29771, DriftBench 2604.28031, ContextEcho 2605.24279, Ross and Lo 2604.23837, Machine Spirits 2604.18602, Hashimoto 2510.12189, ABIDES 1904.12066, Jiang/Peng/Yan JFE 2024, Deflanderization 2510.13586.
19. Delete the section 3.1 claim that realism reintroduces subjectivity. It is false and checkable.

Tier 1 - Construct repair, 4-6 weeks

20. Resolve the context-accumulation question and correct Eq. 3 and Eq. 5 to match the implementation, or reframe the paper as prompt composition (recommended).
21. Fix the P/E leakage, or retract 'hidden fundamental value' and reframe as derivable ground truth.
22. Redesign RG as mandate-conditional regret with an explicit trivial-policy floor. This is a correctness fix, not an improvement.
23. Replace the degenerate $C_{ideal} = 1.0$ target so that perfect adherence is not zero trades.
24. Rename the Caricature Index to Maximum Drawdown and demote it, or build a real trait-exaggeration measure.
25. Retract or re-derive the 4.4x on a non-cumulative statistic with a trend test.
26. Retitle; remove 'psychometric stability'.

Tier 2 - The causal controls the placebo does not provide, 4-6 weeks

27. Four-arm placebo at full grid: static, declarative placebo, directive placebo (length-, delimiter- and imperative-matched but behaviourally irrelevant), and mandate. Report per-persona tests. This is the real answer to the confound objection.
28. Swapped-mandate arm: ISFJ mandate into an ENTJ agent and vice versa. Settles the content-attribution question. 15-90 simulations.

29. Forked restatement probe (knows-but-violates). Approximately 5% extra API calls. The only experiment that can distinguish salience decay from knows-but-violates, and salience decay is the paper's central construct.
30. Mandate wording dose-response: paraphrase, abbreviation, delimiter on and off, imperative strength.
31. Increase seeds to at least 20 in the crash; add at least 3 decode replicates per cell on a subset; fix the temperature misstatement.

Tier 3 - Coverage and method, 3-4 weeks

32. Extend open-weight behavioural coverage to the 30-70B tier with the full metric suite; foreground the Qwen dissociation.
33. Adaptive re-grounding controller with a behavioural trigger (absolute C_t minus C_{ideal} over a threshold). Requires no activations and works on all 18 models. Benchmark explicitly against CLQT's $C_t < 0.7$ design and Meta's fixed-interval schedule.
34. Full-grid O3 numerical-only ablation and continuous OCEAN trait sampling, reframed as vocabulary manipulation rather than validation.
35. Sensitivity on the $EPS = V/15$ constant (10, 15, 25). If model rankings move, RG is measuring P/E-prior alignment.
36. Classifier de-circularisation: state training provenance, retrain on held-out models and arms, add a judge-free surface.

Tier 4 - Mechanism, only if a real effect survives Tiers 0-3

37. M2 rescoped: validate the adherence direction against the restatement-probe label, not against MAS or the classifier. Test whether the internal signal predicts violation better than the observable cash fraction.
38. M3 last, pre-registered, with a random-donor control, self-patch sanity check, matched pairs and seeds, and a stated commitment to publish a null.

Cut for this cycle

M1 (unrunnable as specified; twice published where runnable). M4 (double-gated on a phenomenon that is not significant at model level). The ABIDES port (retain only the argumentative retraction). Full-grid OCEAN scale-up framed as validation.

Part IX - Factual Error and Inconsistency Register

Compiled from Reviewer A's audit. Items marked CONFIRMED were independently re-checked by the orchestrator against the paper text or repository source.

#	Error	Status
1	Figure 5's crash series is sign-inverted relative to Table 11 for all four plotted models; Figure 5 also annotates Gemini 2.5 Pro as 'universal benefit' while plotting its crash bar negative	CONFIRMED
2	Section 4.4 body text uses Figure 5's wrong sign for Qwen (+59.1% instead of -59.1%)	CONFIRMED
3	Table 1 and Table 9 report opposite signs for the same two quantities; Table 10 introduces a third convention	CONFIRMED
4	Figure 3's flat panel spans 0.32-0.36 but Tables 1/9 report full-sample static MAS of 0.391 - arithmetically impossible	Reported
5	Figure 3's bull-trap panel spans 60-80 but Table 1 reports 85.94, above the entire plotted range	Reported
6	Section 4.3.2 claims the bull trap stabilises; Appendix A specifies a blow-off top with exponential disconnection and a 5x volume surge	CONFIRMED
7	Section 4.3.1 explains the bull-trap result by suppressed buying of undervalued assets; Appendix A guarantees the asset is never undervalued there	CONFIRMED
8	Claude Sonnet 4.6 flat/static MAS takes three irreconcilable values across Appendices E, F and H (0.903 / 0.962 / 0.988) while ENTJ static is identical to three decimals at T = 200 and T = 800	Reported
9	Appendix F's stated placebo range 0.003-0.045 should be 0.003-0.047	CONFIRMED
10	Broken self-reference: section 4.4 points to itself instead of Appendix K	CONFIRMED
11	Undefined dagger marker on Gemma-3-4B in Table 11	CONFIRMED
12	RG's stated range of 0.5 to 1.0 is wrong; nothing bounds it below at 0.5 and the random baseline is 2/3	CONFIRMED
13	Ht's max(0,.) clamp is vacuous	CONFIRMED
14	Appendix B.2 instructs ISFJ to prefer index funds and bonds while C_ideal = 1.0 demands 100% cash	CONFIRMED
15	Appendix B.2 instructs ISFJ to buy puts and hedges; the action space is BUY/SELL/HOLD on a single asset	CONFIRMED
16	Table 8 recommends k = 1 for ENTJ while Appendix H's own text states k = 1 worsens ENTJ's MAS from 0.202 to 0.329	CONFIRMED
17	Appendix J's memory benefit is largest in the mildest crash, inverting the stress-accumulation mechanism, and is not discussed	Reported
18	Appendix J's default-discount numbers differ from Tables 1/9 (gap 16.5% versus 12.6%) with no reconciliation	Reported
19	Section 4.4 claims flagships consistently benefit; Table 11's bull-trap column shows GPT-4o - 11.6%, GPT-4.1 -7.4%, GPT-5.4 -8.1%	CONFIRMED
20	Appendix D states temperature 0.0; the code sets 0.2 in three files	CONFIRMED
21	Section 4.3.2 says 'cumulative'; Figure 3's axis says 'rolling' - materially different statistics, and	Reported

	the headline depends on which	
22	Model naming inconsistent: DeepSeek-Chat / deepseek-chat / DeepSeek V3 Chat, cited to a DeepSeek-V3.2 reference	Reported
23	Appendix L is falsified internally: Sonnet 4.6's static Q3 LCR of 67.1% exceeds the mini models' post-injection 44.4% yet shows no reversal; GPT-5-Mini does not reverse and is silently excluded	Reported
24	Revision plan is written against a superseded draft: targets a 4.1x claim (paper says 4.4x), a Gemini 2.5 Flash mislabel (already corrected), Table 7 (now Table 11), and N = 210 / 14 models (now N = 270 / 18)	CONFIRMED

Part X - Open Questions for the Next Review Cycle

Carried forward deliberately. These were not resolvable within this cycle and should be put to the next set of reviewers.

39. Is the architecture as specified in Eq. 3 and Eq. 5, or does the implementation pass portfolio state and history that the formal specification omits? Appendix L rationales reference specific holdings, and portfolio state is absent from Table 2's definition of the observation space. This determines whether the paper's formal specification is wrong or its mechanism claim is.
40. Does a newer draft exist in which the 4.4x fix, the Figure 4/5 captions, or the reframed MBTI defence have landed? Several findings hinge on wording that may already have moved on the authors' side.
41. Which arms supplied the DistilBERT classifier's training data? If memory-arm rationales are included, the P(intended persona) contrast approaches a tautology.
42. Were the three personas ever behaviourally distinguishable at $t = 0$? Ross and Lo's heuristic-collapse result makes this a live question, and the data to answer it already exist.
43. If the swapped-mandate arm shows effects do not track injected content, the paper's thesis changes from mandate semantics to generic instruction salience. Is that reframing acceptable to the authors, and is it still publishable? Reviewer C argues yes, and that it is better learned now than in review.
44. Does the crash effect survive at model level under any defensible specification, or must the panic-selling failure mode be withdrawn entirely?

Appendix A - Reference Corpus

47 papers, all read in substance by at least one reviewer. Locations:

- references/related_work_2026/ - 28 papers (behavioural drift, persona fragility, financial benchmarks, memory mechanisms, governance)
- references/mech_interp/ - 8 papers (persona vectors, refusal directions, attention sinks, instruction instability, ROME, context rot, monosemanticity, lost-in-the-middle)
- references/gap_sweep_aug2026/ - 11 papers plus MANIFEST.md (second sweep; none cited by the paper)

Nearest neighbours, by axis:

Axis	Papers
Goal and mandate drift	Arike 2505.02709; Menon 2603.03258; Drift No More 2510.07777; Everitt 2504.11844
Attention and mechanism	Kim/Li COLM 2024; When Attention Closes 2605.12922; Abdelnabi 2406.00799
Re-grounding interventions	Meta Proactive Memory 2607.08716; CLQT 2606.29771 s3.5; ContextEcho 2605.24279
Knows-but-violates	DriftBench 2604.28031; CLQT coherence gap; Menon say/do gap
Synthetic fundamental markets	Hashimoto 2510.12189; Machine Spirits 2604.18602; ABIDES 1904.12066

Persona and psychometric validity	Kocielnik 2606.12730; Song 2509.10078; PERSIST 2508.04826
Financial agent benchmarks	CLQT 2606.29771; Ross and Lo 2604.23837; KTD-Fin 2605.28359; AI-Trader 2512.10971; Agentic Trading survey 2605.19337
Allocation grounding	Jiang, Peng and Yan, JFE 2024

--- End of report ---